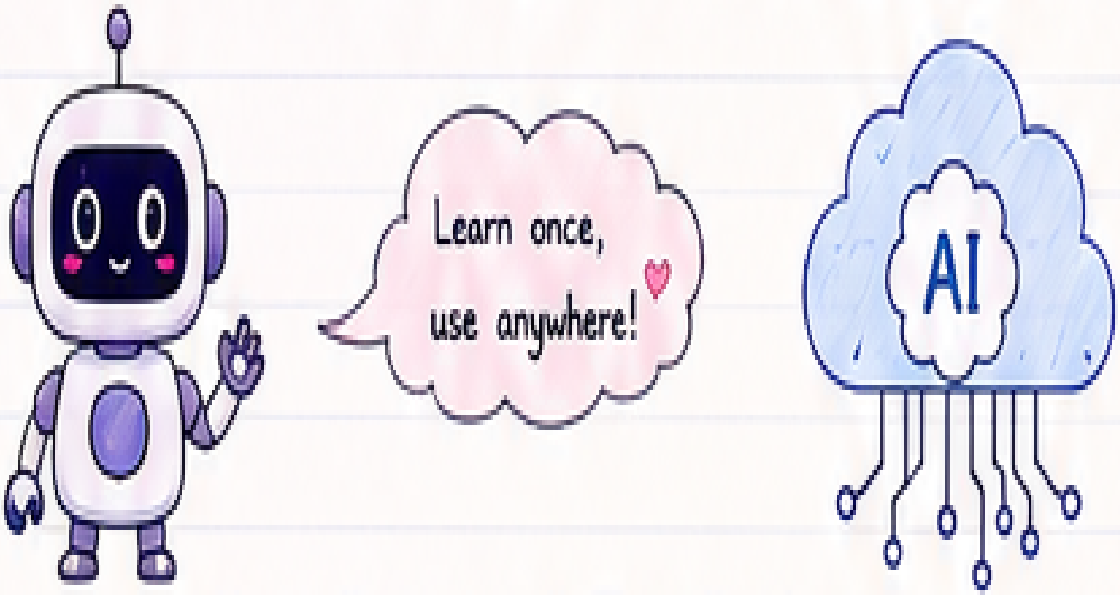
Pre-Trained Networks and Transfer Learning

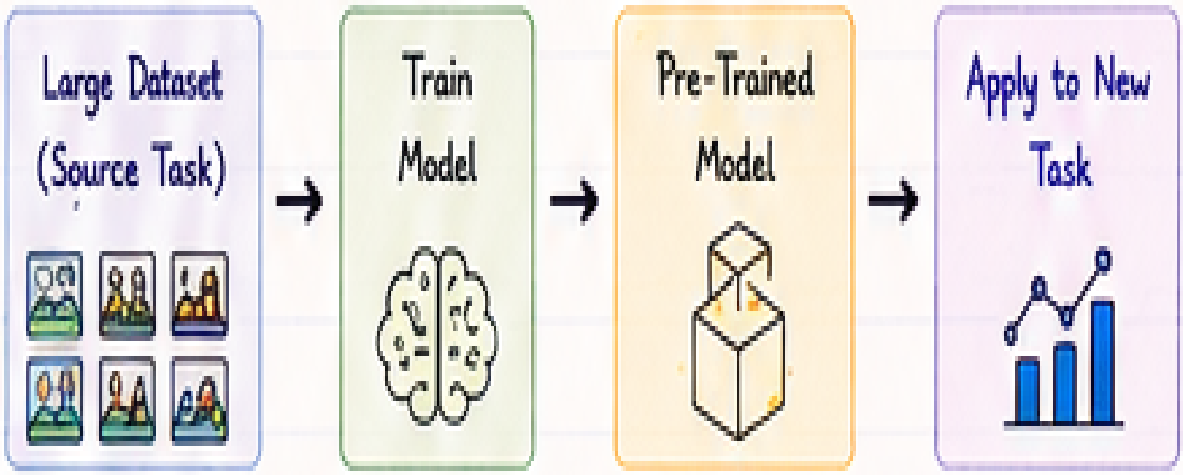
"What is Transfer Learning?"

Transfer Learning is a deep learning technique where a model trained on a large dataset is reused for a new related task, saving training time and improving performance.

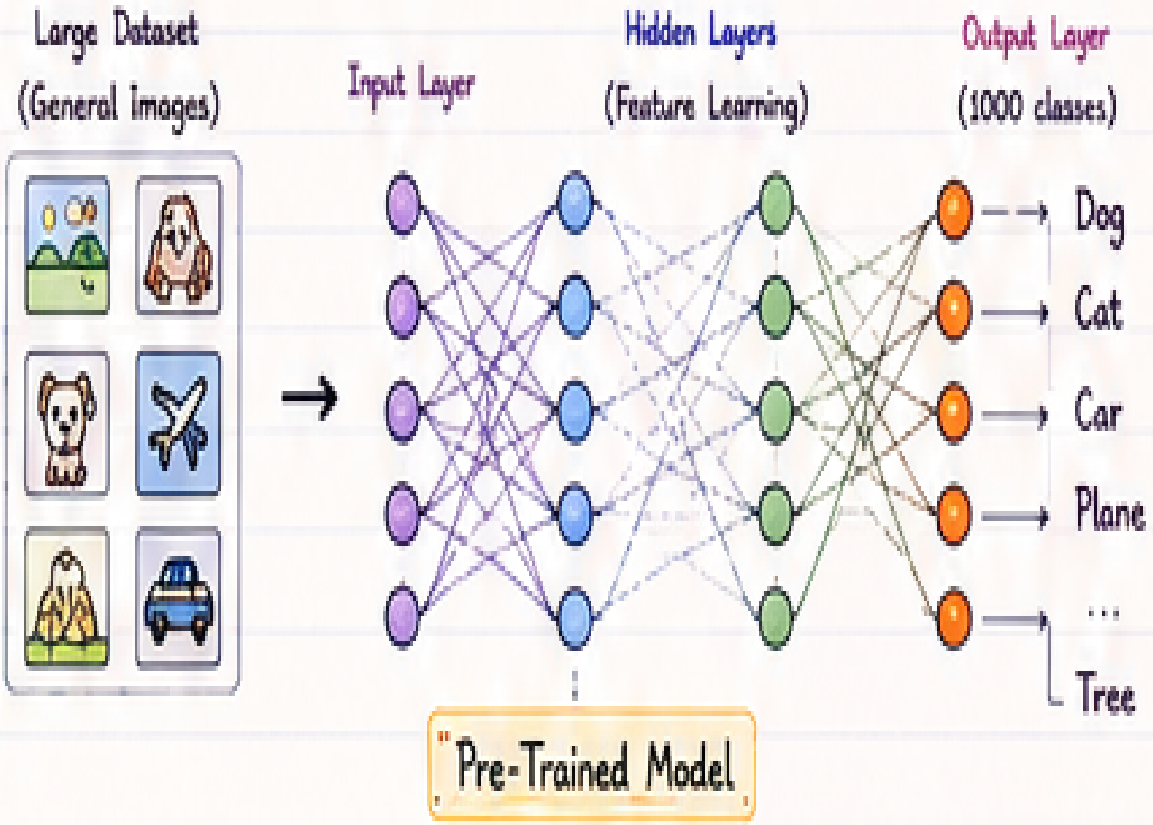


Basic Concepts of Transfer Learning

How Does Transfer Learning Work?



1) "Pre-Trained Model (Source Task)"



Aspect	Traditional Deep Learning	Transfer Learning
Training Method	Train a model from scratch on the target task.	Use a pre-trained model and adapt it to a new task.
Dataset Requirement	Requires a large labeled dataset.	Works well with small labeled datasets.
Training Time	High (long training time)	Low (faster training)
Purpose	Learn all features and patterns from scratch.	Reuse learned features and fine-tune for new task.
Popular Models	• CNN • Custom Neural Network	• VGG16 • ResNet50 • MobileNet • EfficientNet

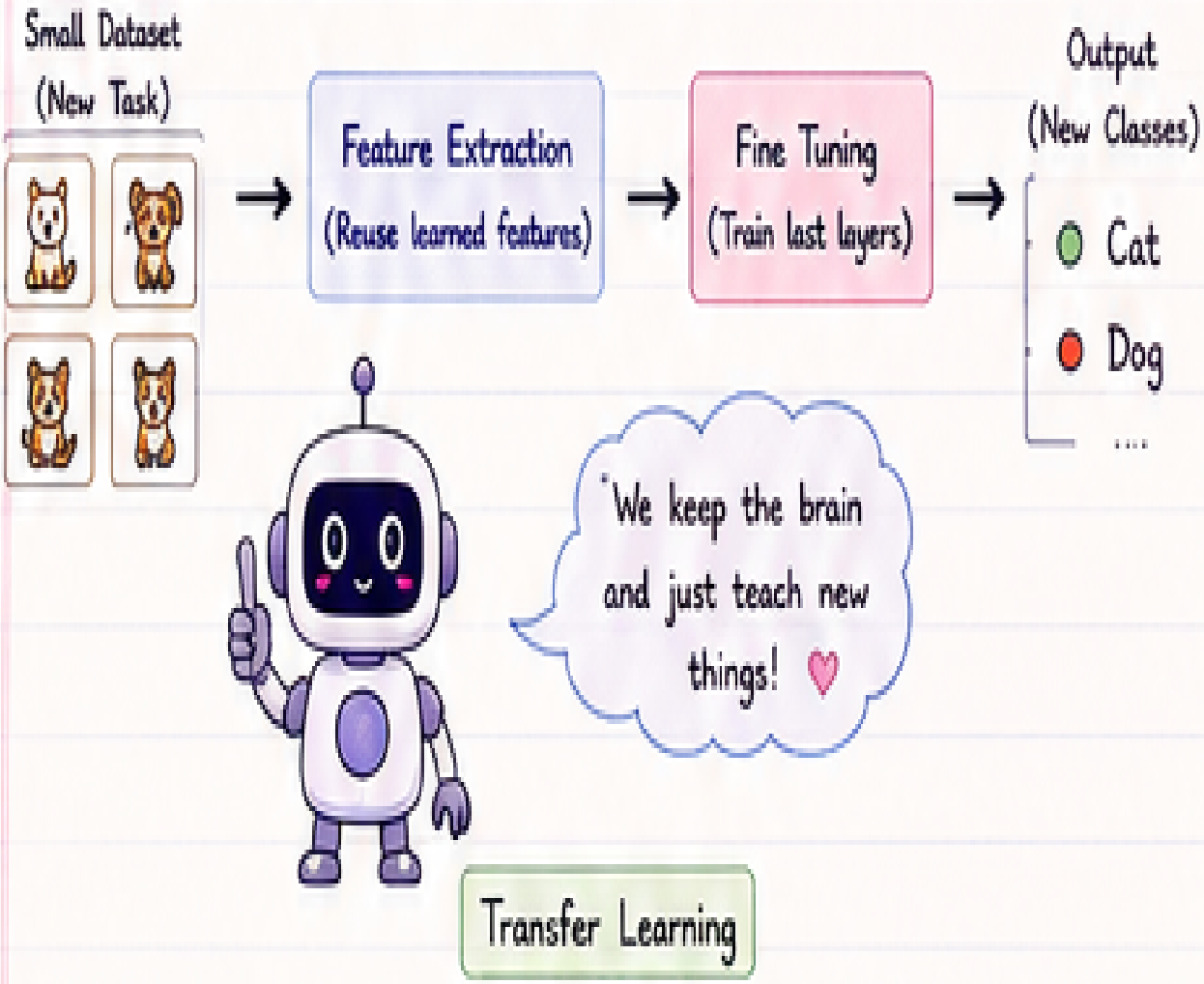
Note: Transfer learning leverages the knowledge gained from solving one problem and applies it to a different but related problem. ♡

Key Components

1. Feature Extraction

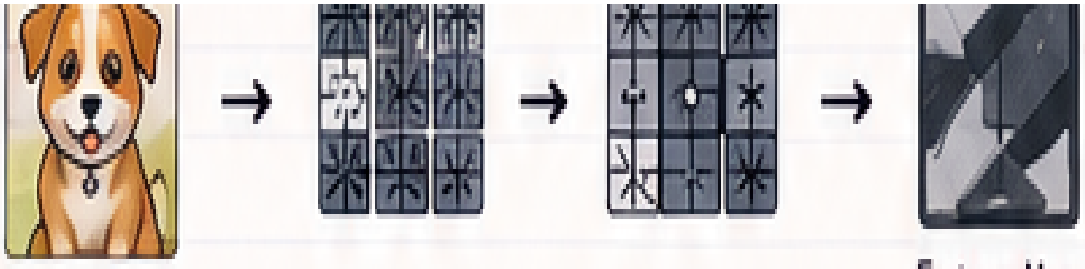
The pre-trained model acts as a feature extractor.
It learns generic features like edges, textures, shapes, etc.

2) Transfer to New Task (Target Task)



Popular Pre-Trained Models

Model	Description	Best For	Input Size
VGG16	16-layer CNN model trained on ImageNet (14M+ images, 1000 classes)	Image classification, feature extraction	224 × 224



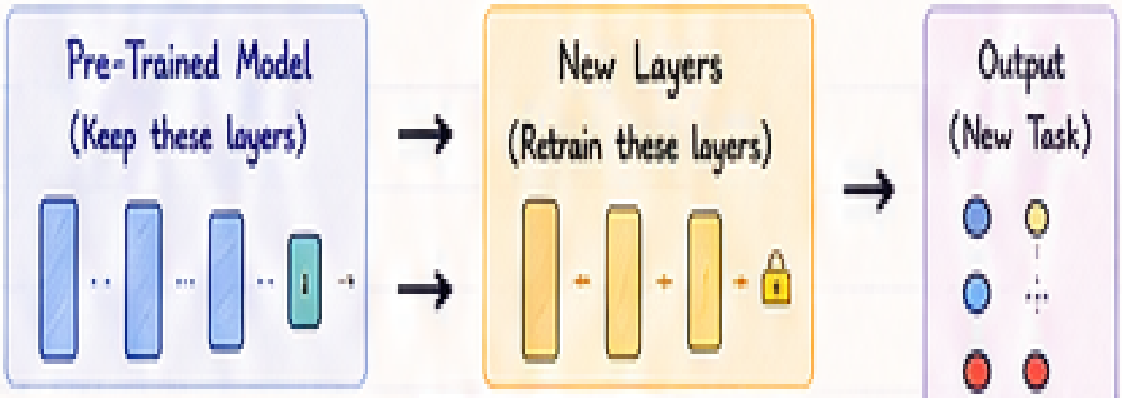
Input Image

Feature Maps
(Higher level features)

" Feature Extraction

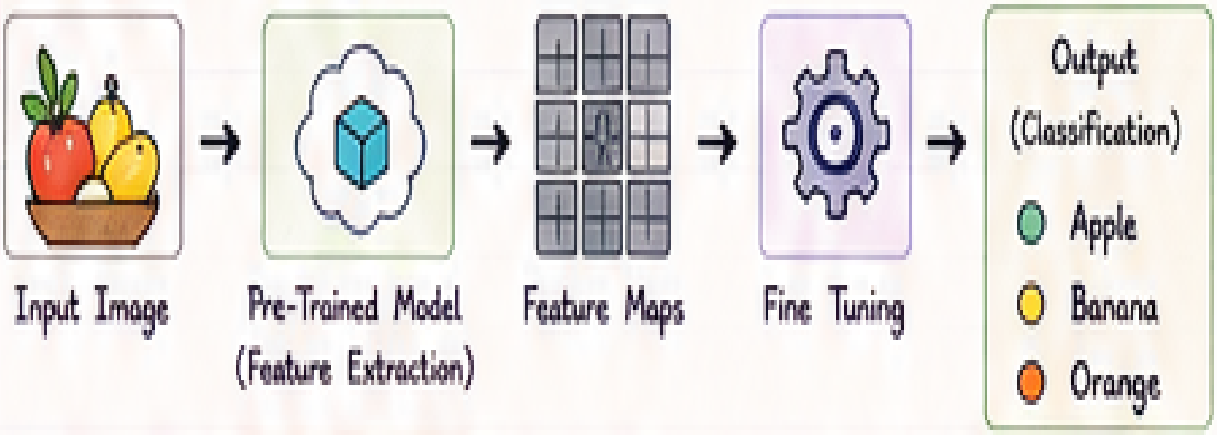
2. Fine Tuning

The last few layers are retrained on the new dataset to adapt to the new task.



Fine Tuning

3. Workflow Summary

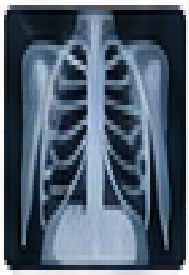


" Transfer Learning = Feature Extraction + Fine Tuning ♡

ResNet50	50-layer deep residual network, solves vanishing gradient problem	Complex image tasks, fine-grained tasks	224 × 224
MobileNet	Lightweight model for mobile & edge devices	Mobile apps, real-time applications	224 × 224
EfficientNet	High accuracy with fewer parameters using compound scaling	High performance with less computation	224 × 224

Applications of Transfer Learning

Medical Imaging



Detect diseases from scans

Self-Driving Cars

Object detection & navigation

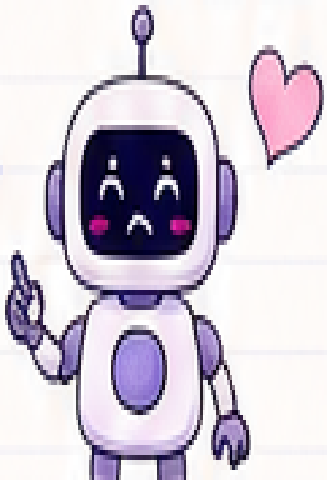
Facial Recognition

Identify & verify individuals

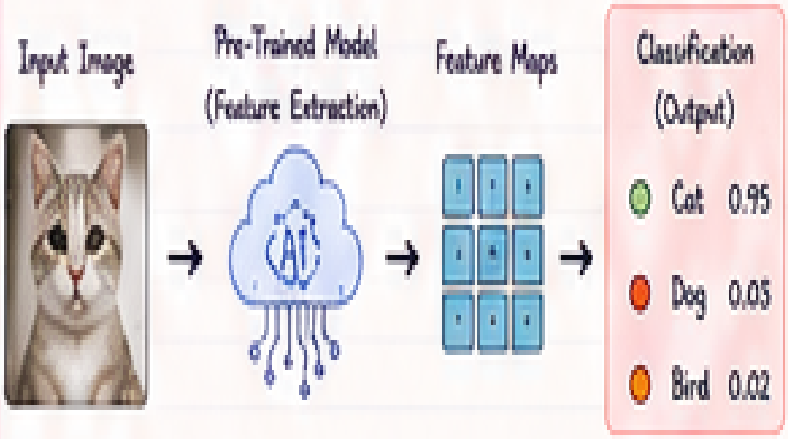
Natural Language Processing

Improve text models with less data

"Transfer learning helps us build smart models faster and with less data!



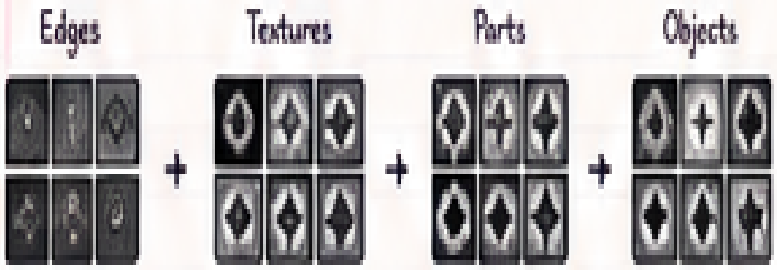
Example: Image Classification

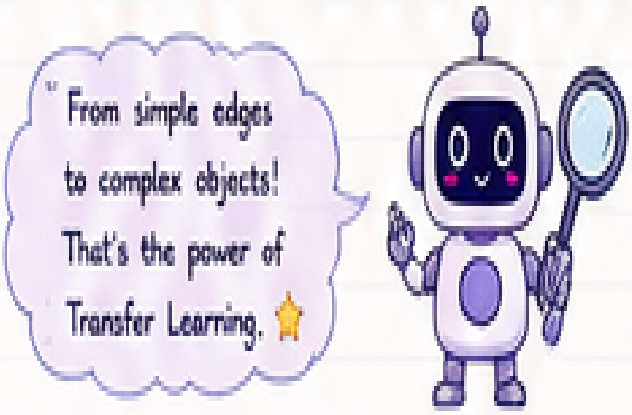


What happens inside?

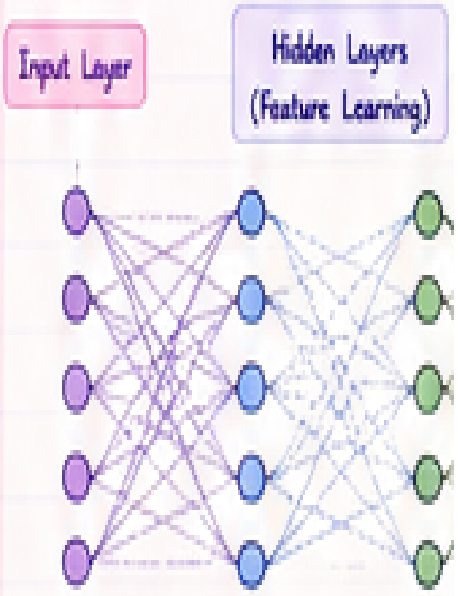
- The input image passes through the pre-trained model.
- The model extracts important features (edges, shapes, textures).
- Final layers classify these features into target classes.

Feature Visualization (Example)





Neural Network Layers



Early Layers (General Features)

- Detect edges, colors, simple shapes
- Learned from large datasets

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Benefits of Transfer Learning

- ✅ Reuses existing knowledge
- ✅ Requires less training data
- ✅ Faster convergence
- ✅ Better performance with limited dat
- ✅ Reduces computational cost